

LIBRARY MANAGEMENT SYSTEM

SEARCH ALGORITHMS:

i)Linear Search

ii)Binary Search

Linear Search

Linear Search checks each element one by one until the desired element is found or the list ends.

Algorithm

1. Start from the first element.
2. Compare the current element with the target.
3. If matched, return the position.
4. Otherwise move to the next element.
5. Repeat until the element is found or the list ends.

Time Complexity

- Best Case: $O(1)$
- Average Case: $O(n)$
- Worst Case: $O(n)$

BINARY SEARCH

Binary Search works only on a sorted list. It repeatedly divides the search space into two halves.

Algorithm

1. Find the middle element.
2. Compare the middle element with the target.
3. If equal, return the position.
4. If target is smaller, search the left half.
5. If target is greater, search the right half.
6. Repeat until found.

Time Complexity

- Best Case: $O(1)$
- Average Case: $O(\log n)$
- Worst Case: $O(\log n)$